

MINERALS

Minerals are the materials from which rocks are made. More than 4,000 different minerals exist, each one with its own chemical composition, found naturally on Earth. Many minerals are hard and crystalline, and, while some are extremely abundant, others – including diamond – are very rare, and highly prized.



Copper often appears in a dendritic form, branching like a tree. It is a very important economic mineral.



Malachite can be botryoidal, with small, rounded shapes, or massive, with no definite shape.



Crocoite, which is lead chromate, frequently occurs as slender, elongated prismatic crystals.

Minerals are of great economic significance. They provide us with countless useful materials, from metals to industrial catalysts, and they also include objects of great intrinsic beauty, particularly when cut and polished as gems. But on an even broader scale, minerals are essential for life itself. In soil and in water, soluble minerals release a steady stream of chemical nutrients that plants and other organisms need to grow. Without them, the world's ecosystems would no longer operate.

Minerals are classified according to their chemistry. A few minerals, such as gold, silver, and sulfur, can exist in a native state, meaning they contain a pure chemical element, and nothing else. All other minerals are chemical compounds. Quartz, for example, contains two elements – silicon and oxygen – which are bound together very tightly, giving quartz its exceptional hardness and strength. Under the Strunz classification system, quartz (silicon dioxide) is classified as an oxide, as are chalcedony and opal, but under the Dana system, used in this book, they are included with the silicates. The largest group of minerals, the silicates make up about 75 per cent of Earth's crust. Other common mineral groups include sulfides, oxides, carbonates, arsenates, and halides.

IDENTIFYING MINERALS

With experience, many minerals can be identified by their appearance alone. Important clues include their colour, their lustre – the way light is reflected from their surfaces – and above all, their habit, and/or crystal form. Crystals are grouped into six systems according to their symmetry (see panel, below). Minerals do not always occur as crystals but can take different forms: they can be dendritic (branching), or botryoidal (like a bunch of grapes), for example.

Minerals also differ in density, or specific gravity (SG) – measured by comparing the weight of a mineral to that of an equal volume of water – and in their hardness (H). On the 10-point Mohs scale, which measures hardness, talc is rated 1, whereas diamond, the hardest mineral, is rated 10. A fingernail (hardness 2½), copper coin (3½), and steel knife blade (5½) all make useful benchmarks for testing hardness. Surprisingly, size is not a useful clue. Gypsum crystals, for example, are usually less than 1 cm (¾ in) long, but the largest specimens ever discovered are the size of a two-storey house.

VOLCANIC MINERALS >

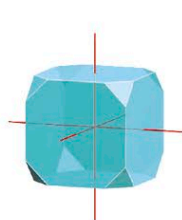
The ground at Dallol in Ethiopia's Danakil Desert is pock-marked by volcanic vents and covered in sulfur, a native element.

DEBATE

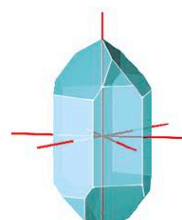
MINERALS OR NOT?

Minerals are traditionally considered to be inorganic. Although some organic materials are used as gemstones and for other decorative purposes, and some of these have a similar chemical composition to minerals, they are not true minerals. Amber is the hardened resin from trees; much amber is of Cretaceous, Palaeogene, and Neogene age and can contain fossilized insects. Jet, a soft black rock, is a type of impure coal, and became highly fashionable for jewellery in Victorian times. Shell, pearl, and coral (the use of which is not sustainable) are rich in calcite and are used decoratively.

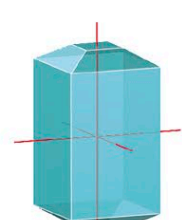
CRYSTAL SYSTEMS



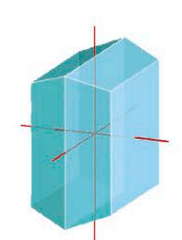
Crystals in the **cubic** system are relatively common and easily recognized. Such crystals have three equal axes at right angles; shapes include the cube and the eight-faced octahedron.



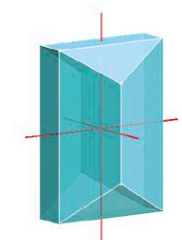
Hexagonal and trigonal systems are very similar to each other, with four axes of symmetry. Their crystals are often six-sided prisms with pyramidal tops (left).



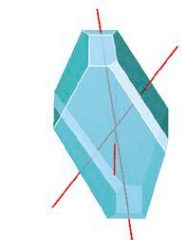
Crystals in the **tetragonal** system have three axes of symmetry, all at right angles. In a tall prism (left), the vertical axis is the longest. Squat prisms are also common.



Crystals in the **monoclinic** system have three unequal axes of symmetry, only two at right angles. Tabular (flattened, left) and prismatic habits are common.



Crystals in the **orthorhombic** system are similar to those in the monoclinic system, but all three axes are at right angles; the usual habits are tabular and prismatic (left).



Crystals in the **triclinic** system have a low degree of symmetry, as the three axes are unequal in length, with no right angles between them. Prismatic habits are common.